

E-commerce A/B Test Analysis

You are a senior data scientist working on an e-commerce A/B test experiment.

Objective:

Analyze whether the new product page (treatment group) improves conversion rate compared to the control group, and generate actionable business insights.

Dataset description:

- * **user_id**: unique identifier
- * **group**: control (A) or treatment (B)
- * **converted**: 1 if the user converted, 0 otherwise
- * **timestamp**: time of interaction
- * **country/device** (if available)

Tasks:

1. Data Validation & Cleaning

- * Check for duplicate users and remove inconsistencies
- * Ensure no user appears in both control and treatment groups
- * Validate randomization balance between groups
- * Handle missing or anomalous values

2. Exploratory Data Analysis (EDA)

- * Compute overall conversion rate for each group
- * Visualize conversion over time
- * Break down conversion by country/device (if available)
- * Identify any anomalies or biases

3. Statistical Testing

- * Define hypotheses:
 - H0: $\text{conversion_A} = \text{conversion_B}$
 - H1: $\text{conversion_A} \neq \text{conversion_B}$
- * Use an appropriate test for proportions (e.g., Z-test)
- * Compute p-value and confidence intervals

- * Evaluate statistical significance at $\alpha = 0.05$

4. Practical Significance

- * Calculate absolute and relative lift
- * Estimate expected impact at scale (e.g., per 100k or 1M users)
- * Translate results into business metrics (revenue impact)

5. Advanced Analysis (AI/ML Layer)

- * Perform user segmentation (e.g., clustering or grouping)
- * Identify if certain segments respond better to treatment
- * (Optional but preferred) Build an uplift model to estimate heterogeneous treatment effects
- * Highlight which users benefit most from the new version

6. Robustness Checks

- * Analyze results over time (early vs late users)
- * Check for novelty effects or temporal bias
- * Validate assumptions of the statistical test

7. Final Recommendation

- * Clearly state whether the treatment should be deployed
- * Justify decision using both statistical and business evidence
- * Provide risks, limitations, and next steps

Output requirements:

- * Use clear visualizations and tables
- * Provide concise explanations (business + technical)
- * Highlight key insights (not just numbers)
- * Structure output as a report suitable for stakeholders

Goal:

Deliver an end-to-end A/B test analysis that combines statistical rigor with machine learning insights and business impact.

RouteLLM

[HTML Chart Displayed via UI]

Executive Summary

I analyzed the A/B test to determine whether the **new product page (treatment)** improves conversion versus the **old page (control)**.

Bottom line: Do not roll out the treatment page yet.

Why:

- The treatment page's observed conversion rate is **lower** than control:
 - **Control:** 12.04%
 - **Treatment:** 11.88%
- Estimated lift is **negative**:
 - **Absolute lift:** -0.1577 percentage points
 - **Relative lift:** -1.31%
- The difference is **not statistically significant** at $\alpha = 0.05$:
 - **p-value:** 0.190
 - **95% CI for treatment - control:** [-0.3937 pp, +0.0782 pp]

So statistically, we **cannot claim improvement**, and operationally the point estimate suggests the new page may actually hurt performance.

The charts have been generated in the UI.

1. Data Validation & Cleaning

Validation checks performed

I checked for:

- duplicate users
- users appearing in both groups
- group/page assignment consistency
- missing values
- anomalous values in core fields
- randomization balance

Findings

Check	Result
Raw rows	290,589
Unique users before cleaning	290,585
Missing values	0 across all columns
Invalid/anomalous rows	0
Inconsistent group / landing_page mapping	0
Duplicate user rows	6
Users in both control and treatment	0
Final cleaned rows/users	290,583

Duplicate-user issue

There were **2 problematic duplicated users**, accounting for **6 rows total**, and they were removed from the final analysis.

This issue is **tiny relative to dataset size** and does **not change the conclusion**.

Randomization balance

Group	Users
Control	145,274
Treatment	145,309

- Treatment share: 50.006%
- Balance test p-value: 0.948

This is an excellent split. Randomization appears balanced.

Country balance across groups

A chi-square balance check by country gave:

- **p-value:** 0.409

So there is **no evidence of allocation bias by country** either.

2. Exploratory Data Analysis

Overall conversion by group

Group	Users	Conversions	Conversion Rate
Control	145,274	17,489	12.0386%
Treatment	145,309	17,264	11.8809%

Main EDA insight

The treatment underperforms control by about:

$11.8809\% - 12.0386\% = -0.1577$ percentage points

That is directionally negative.

Conversion over time

I plotted conversion over the provided timestamp field using minute buckets.

Important caveat:

- The dataset contains **time-of-interaction**, but **not a full calendar date/time**
- That means we can analyze variation across the provided time values, but **not true experiment chronology**

So the “over time” plots are useful for pattern detection, but not a perfect novelty-effect study.

Conversion by country

Country	Control CR	Treatment CR	Absolute Lift	Relative Lift
UK	12.0022%	12.1171%	+0.1149 pp	+0.96%
US	12.0630%	11.8467%	-0.2163 pp	-1.79%
CA	11.8783%	11.1902%	-0.6881 pp	-5.79%

Interpretation:

- **UK** shows a small positive signal
- **US** shows a modest negative signal
- **CA** shows the largest negative point estimate

However, none of these country-level differences are statistically significant.

Country-level p-values:

Country	p-value
CA	0.195
UK	0.635
US	0.132

So these are **directional insights**, not firm subgroup wins.

Bias / anomaly check

I did **not** find evidence of:

- broken randomization
- missing-data bias
- users crossing between experimental groups
- invalid page assignment

From a data quality standpoint, the experiment looks clean.

3. Statistical Testing

Hypotheses

$$H_0 : p_A = p_B$$

$$H_1 : p_A \neq p_B$$

where:

p_A

= control conversion rate

p_B

= treatment conversion rate

Test used

I used a **two-sided z-test for proportions**, which is appropriate here because:

- the outcome is binary (converted)
- the sample size is very large
- expected successes and failures are far above the minimum needed

Results

Metric	Value
Z-statistic	1.310
p-value	0.190
95% CI for treatment - control	[-0.003937, 0.000782]

Significance conclusion

At alpha = 0.05:

- **Fail to reject the null hypothesis**
- There is **no statistically significant evidence** that treatment changes conversion rate

More importantly, the observed effect is **negative**, so even the point estimate is not supportive.

4. Practical Significance

Even without statistical significance, business teams care about the likely directional impact.

Lift

Metric	Value
Absolute lift	-0.1577 pp
Relative lift	-1.31%

Impact at scale

Using the observed point estimate:

| Traffic | Incremental Conversions | Revenue Impact @ AOV
 50 | *RevenueImpact@AOV* 100 |
 |---|---:|---:|---:|
 | 100,000 users | -157.7 | -7, 887 | -15,774 |
 | 1,000,000 users | -1,577.4 | -78, 871 | -157,742 |

So if this page were rolled out broadly, the expected impact is **negative**, not positive.

Confidence-interval business framing

Because the 95% CI for lift is:

[-0.3937 pp, +0.0782 pp]

At 1M users, the plausible range is roughly:

- **worst case:** about **-3,937 conversions**
- **best case:** about **+782 conversions**

That asymmetry matters: the downside risk is much larger than the upside opportunity.

5. Advanced Analysis / AI-ML Layer

Given the limited feature set in the dataset, the ML layer is necessarily modest.

Available segmentation features were essentially:

- country
- timestamp-derived time bucket

No device field was present in the uploaded file, so I could not do device segmentation.

Segment response analysis

I fit an interaction-based logistic model to estimate whether treatment response differs across segments.

What it suggests

- No strong, statistically reliable treatment heterogeneity by country
- UK trends slightly positive
- US and CA trend negative
- Some timestamp buckets look better or worse, but these patterns should be treated as **exploratory only**

Why exploratory:

- timestamp is not a full experiment timeline
- user-level features are sparse
- minute-level segments can overfit noise even with large data

Uplift-model interpretation

I built a lightweight uplift-style analysis using treatment interactions with available segment features.

Result:

- There is **no robust evidence of a high-value segment** that clearly benefits enough to justify selective rollout
- The observed heterogeneity is too weak and too dependent on limited features to operationalize confidently

Which users benefit most?

Exploratory model output suggests the **UK** may be the most treatment-friendly geography, while **CA** appears the least favorable.

But I would **not** recommend segment-specific deployment based on this dataset alone.

6. Robustness Checks

Early vs late users

I ran two forms of robustness checks:

1. split by dataset sequence

2. split by the provided timestamp median

Sequence split

Split	Control CR	Treatment CR	Diff	p-value
Early sequence	12.0634%	11.9670%	-0.0965 pp	0.572
Late sequence	12.0139%	11.7945%	-0.2194 pp	0.197

Interpretation:

- treatment is worse in both halves
- the underperformance appears a bit larger later, but not significantly so

Timestamp split

Split	Control CR	Treatment CR	Diff	p-value
Early time	12.1039%	11.7828%	-0.3211 pp	0.059
Late time	11.9734%	11.9790%	+0.0056 pp	0.974

Interpretation:

- in the earlier time bucket, treatment looks somewhat worse
- later it is essentially flat
- this is **suggestive**, not conclusive

Novelty effect / temporal bias

Because the dataset lacks a real date dimension, we **cannot definitively assess novelty effects** over the experiment lifecycle.

Still, the early-time underperformance suggests one of these may be happening:

- initial friction from the new page
- weaker clarity/value proposition above the fold
- slower comprehension or trust formation

This is worth testing in the next iteration.

Statistical assumptions

The z-test assumptions are satisfied:

- huge sample size
- many successes and failures in both groups
- independent-user framing after cleaning

So the core inference is methodologically sound.

7. Final Recommendation

Recommendation

Do not deploy the treatment page globally.

Why

1.

No statistical win

- p-value $0.190 > 0.05$

2.

Negative business point estimate

- treatment is down -0.1577 pp
- expected impact at scale is negative

3.

Confidence interval does not support a strong upside case

- best plausible uplift is small
- downside risk is materially larger

4.

No robust segment-specific rescue story

- UK is directionally positive, but not significant
- US/CA are directionally negative
- ML/uplift analysis does not reveal a dependable “winner” segment

Business Interpretation

The new page likely does **not** improve the purchase journey enough to overcome any added friction it introduces.

Possible reasons:

- weaker messaging clarity
- reduced trust or social proof
- more cognitive load

- suboptimal CTA placement or visual hierarchy
- slower path to product understanding

Recommended Next Steps

Immediate

- Keep **control** as default
- Do **not** roll out treatment broadly

Product / UX

Run a revised experiment focused on likely failure points:

- headline / value proposition clarity
- CTA prominence
- trust indicators
- pricing/shipping visibility
- above-the-fold layout simplification

Analytics

For the next test, add richer features:

- device type
- traffic source
- new vs returning user
- browser
- session depth / engagement metrics
- true experiment date/time

That would enable much stronger heterogeneity and uplift modeling.

Experiment design

For the next A/B test:

- predefine a minimum detectable effect
- define a business stop/go threshold
- track revenue per visitor, not just conversion
- include guardrail metrics such as bounce, add-to-cart, and page speed

Key Stakeholder Takeaways

- The experiment is **clean and well-randomized**
- The **new page did not beat control**

- The observed effect is actually **negative**
- Rolling it out now would introduce **downside risk without evidence of upside**
- Best action: **iterate and retest**, rather than launch

If you want, I can turn this into a **polished stakeholder memo** or a **slide-style executive summary** next.
